

A Standardized Framework for the Number of Flaps n^2 22.5°-induced Bird Base Compounds Produce when Tiled within a Square

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Abstract: A system of proofs that lead to a standardized formula to evaluate the number of flaps a compound of n^2 bird bases yield on a square. For brevity, the proofs in this paper assume a working knowledge of basic geometric properties, coupled with trigonometric principles. This paper solely analyses the effect of tiling petal-like bird base flaps with a visible area of 10.354% of the total square area. Empirically testing crease patterns on an Origami Simulator for their flat-folded patterns, it concludes with a precise formula for any n^2 bird base tiles.

Keywords: Origami; Trigonometry; Square tiling; Tree theory; Structural design

1 Proof I: 22.5° Angle Division in Squares

Constraining the quadrilateral to the limits of a square, we rationalize four right angles (90°) in each corner.

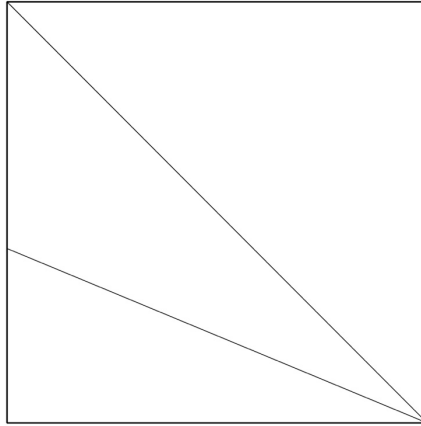


Figure 1

Enforcing the diagonals, we thus bisect the right angle to yield a 45° ($\therefore 90^\circ/2$) angle. We further bisect the 45° angle to achieve a 22.5° ($\therefore 45^\circ/2 = 90^\circ/4$) angle.

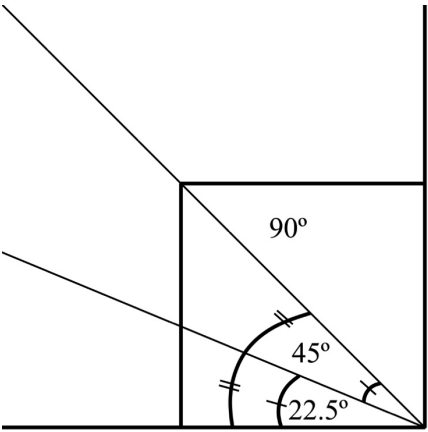


Figure 2

Below, we shall start with a 22.5° -angled scalene triangle.

2 Proof II: A Set of Right-angled Triangles with a 22.5° Co-angle

In this section, we first qualitatively describe the 22.5° -induced flap before mathematically describing its ratios, which shall serve as the foundation for the empiric analysis of the bird base flaps.

2.1 Describing the Right-angled Triangle and its neighbors

In a given square illustrated below, we equate each side to $2u$, such that half of the side measures exactly u . We then restrict the 22.5° -angle-supported line segment running from the base corner to the opposite side, with a parallel line that bisects the base side. This yields a scalene triangle with base u , and opposite angles of

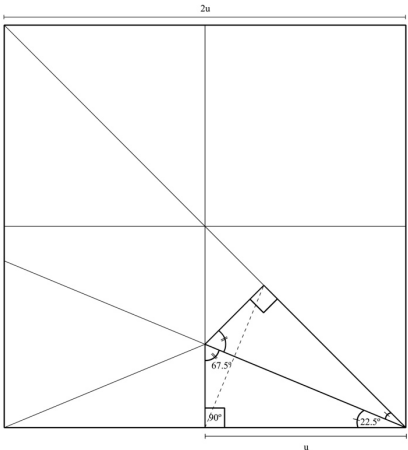


Figure 3

22.5° and 67.5° respectively. This condition is mirrored onto the upper segment divided by the 22.5° incline, implying two triangles that are completely congruent. We now extend our scope to the remaining triangle, a part of the larger diagonal-central bisector-induced isosceles triangle. The smaller triangle is an isosceles triangle, which imitates the property of the larger triangle we have described. Using the Pythagorean formula for the larger isosceles triangle, we can compute that the hypotenuse measures $\sqrt{2}u$ ($\because \sqrt{u^2+u^2}$).

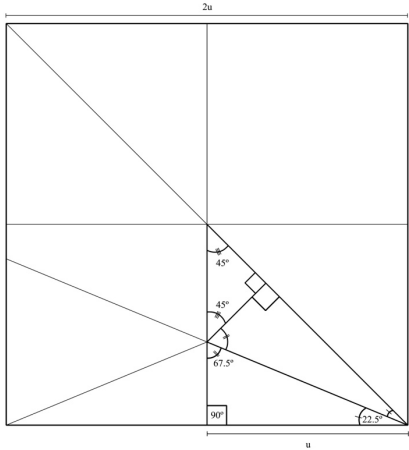


Figure 4

Significant to our final assessment, the highlighted line segments in *Figure 5* are equal, due to the congruence of the two scalene triangles, and then due to the condition of equal sides on an isosceles triangle. Prior to our computation of the ratio necessary to describe the long 22.5°-induced

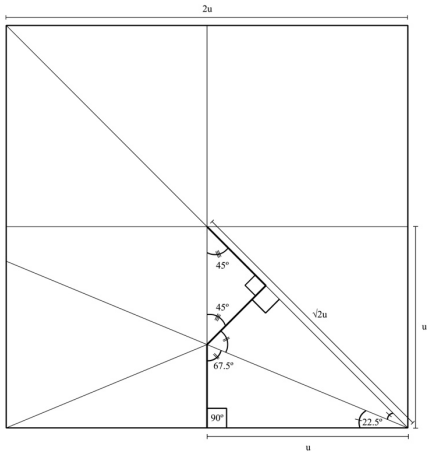


Figure 5

flap, we label the points of our figure, allowing the base $\overline{CA}$ to equal u , while $\overline{EC}$ equals $\sqrt{2}u$. Hence, the sides of the square $ACGE$ are equal:

$$\overline{AE} = \overline{EG} = \overline{GC} = \overline{CA} = u,$$

and

$$\overline{AB} + \overline{BE} = \overline{AE}.$$

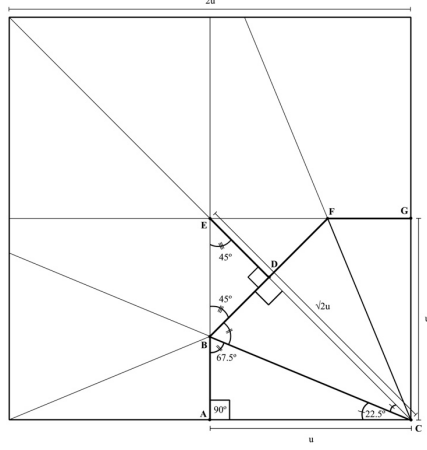


Figure 6

Moreover, our previous statement now implies that:

$$\overline{AB} = \overline{BD} = \overline{DE} = \overline{DF} = \overline{FG},$$

and

$$\overline{BC} = \overline{FC}.$$

2.2 A Mathematical Description of the 22.5°-induced flap ABFGC

Expounding on *Figure 6*, we can apply the Law of Sines to evaluate the length of each side:

$$\begin{aligned} \frac{\overline{AB}}{\sin C} &= \frac{\overline{BC}}{\sin A} = \frac{\overline{CA}}{\sin B} \\ \Rightarrow \frac{AB}{\sin 22.5^\circ} &= \frac{BC}{\sin 90^\circ} = \frac{AC}{\sin 67.5^\circ}. \end{aligned}$$

Assuming

$$\overline{CA} = u = 1,$$

we have:

$$\frac{\overline{AB}}{0.38268} = \frac{\overline{BC}}{1} = \frac{\overline{CA}}{0.92388}$$

$$\begin{aligned}\Rightarrow \overline{AB} &\approx 0.41412 \\ \Rightarrow \overline{BC} &\approx 1.08239.\end{aligned}$$

Since $\overline{AB} = \overline{BD} = \overline{DE}$, we now know that:

$$\overline{AB} = \overline{BD} = \overline{DE} \approx 0.41412.$$

Applying the Pythagorean theorem, we have:

$$\begin{aligned}\overline{BE}^2 &= \overline{BD}^2 + \overline{DE}^2 \\ \Rightarrow \overline{BE} &\approx 0.58565.\end{aligned}$$

We can thus verify our earlier equality assumption:

$$\overline{AB} + \overline{BE} = 0.41412 + 0.58565 = 0.99977 \approx 1 = u = \overline{AE}.$$

This cyclic proof now yields $\triangle BFC$ with an area of:

$$\begin{aligned}&\frac{1}{2} \times \overline{BF} \times \overline{DC} \\ &\Rightarrow \overline{BD} \times (\overline{EC} - \overline{ED}) \\ &\Rightarrow 0.41412 \times (\sqrt{2} - 0.41412) \\ &\approx 0.41412 \times (1.41421 - 0.41412) \\ &\approx 0.41416 \text{ sq. units.}\end{aligned}$$

Since we earlier assigned $u = 1$, we have:

$$\text{Area}(\triangle BFC) = 0.41416u^2.$$

However, the area of the flap is scaled to AC , which is $\frac{1}{2}$ of the square side. Therefore, we can conclude that in ratio to the entire square, $\triangle BFC$ assumes:

$$\frac{0.41416u^2}{(2u)^2},$$

which simplifies to yield approximately 0.10354, or **10.354%**. We have now described the bird base flap that shall be empirically tiled within a square below.

3 An Empiric Assessment of n^2 22.5° Angle Compounds in a Square

A square can only accept a perfect square n^2 ($n^2 \in 1^2, 2^2, 3^2, 4^2 \dots \infty$) tiles of bird bases due to its reliance on the principle of squaring.

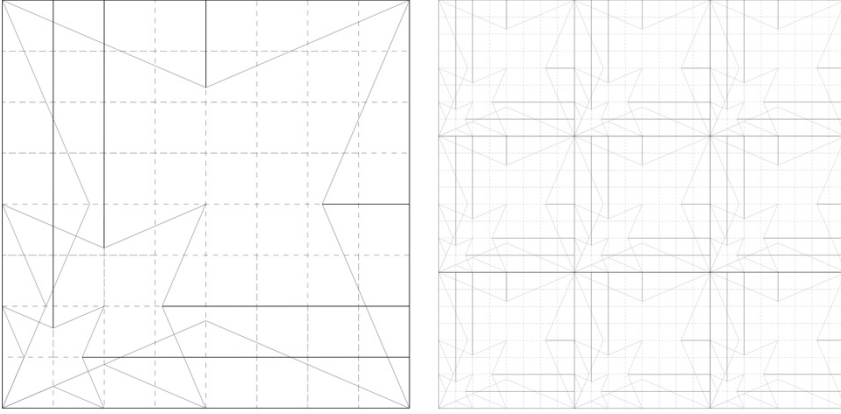


Figure 7

An important clarification regarding the scope of our consideration of flaps must be made before progressing onto the empiric assessment. This paper discards the count of any isosceles central 'connector-flaps' that arise within the bird base and its compounds in its study. Hence, it precisely considers solely 22.5° -induced flap modules, which are the longest structures within a bird base compound. Moreover, the paper 'visible' in these folds, as mentioned in the introduction, possess the area of approximately 10.354% if viewed at once (see *Figure 8*).

4 A Standard Formulaic Framework

4.1 The Empiric Process

Using the Origami Simulator developed by Ghassaei (2017), we can produce flat-folded versions of our iterations of scaled tiles (Lang, 1996) of perfectly squared



Figure 8

bird base compounds that fill an entire square structure. Illustrated below is one bird base, and its implied structure of 22.5° -implied petal structures:

Scaling, we also acquire:

4.2 Formulation

The above section empirically demonstrates a series of possible maps, which is better understood if illustrated:

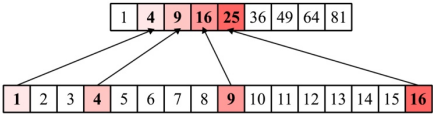


Figure 11

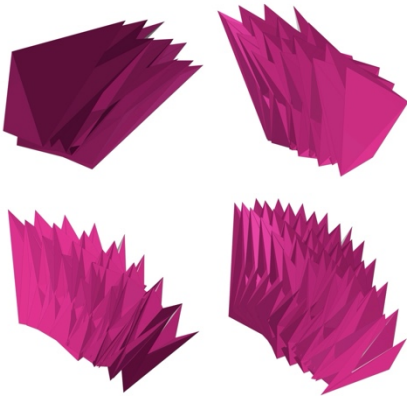


Figure 9

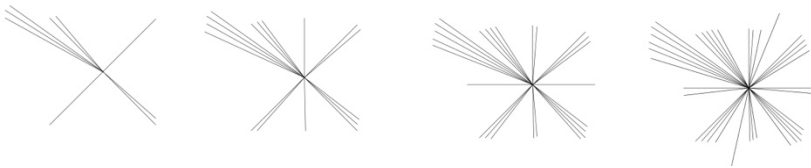


Figure 10

While *Figure 11* does not exhaust the application of this series, it simplifies to a map of squared values:

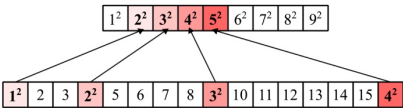


Figure 12

Which may then be simplified to the following map:

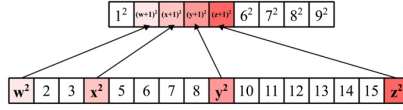


Figure 13

The map describes a function $f(n) = n^2$, such that it corresponds to $f(n) = (n+1)^2 = g(n+1)$.

The formula is thus described as follows:

$$n^2 \text{ bird base tiles} \rightarrow (n+1)^2 \text{ 10.354\% perceived flaps.}$$

To apply this formula, we achieve, for 1 bird base:

$$\begin{aligned} n &= 1 \\ \Rightarrow n^2 &= 1 \\ \Rightarrow (n+1)^2 &= 2^2 = 4. \end{aligned}$$

Which implies that from 1 bird base, we achieve 4 long petal-like 10.354%-area perceived flaps. This may be scaled to match our observations from the empiric assessment:

$$\begin{aligned} 4 &\rightarrow 9 \\ 9 &\rightarrow 16 \\ 16 &\rightarrow 25 \\ 25 &\rightarrow 36, \end{aligned}$$

and so on. It is also significant to note that the effort required for a compound of $n^2 = 25$ already reduces the existing incentive to fold it, and hence if scaled to $n^2 = 100$ may be impractical despite the formulaic accuracy.

4.3 Proof III: Inverse Proof of the Standard Formula

Tiling any deviation from n^2 will disqualify the validity of this framework, while also invalidating the input due to its incomplete coverage of the square. An inverse application of this framework implies that the following must be true:

$$n^2 \text{ bird base tiles} \leftrightarrow (n+1)^2 \text{ 10.354\% perceived flaps.}$$

Which, when applied to a hypothesized set of 49 flaps, is supposed to yield 36 bird bases:

$$\begin{aligned} (n+1)^2 &= 49 \\ \Rightarrow n+1 &= 7 \\ \Rightarrow n &= 6, \end{aligned}$$

and,

$$n^2 = 36.$$

This is verifiable empirically using the Origami Simulator (Ghassaei, 2017) for a crease pattern tiled with 36 bird bases, as it yields:

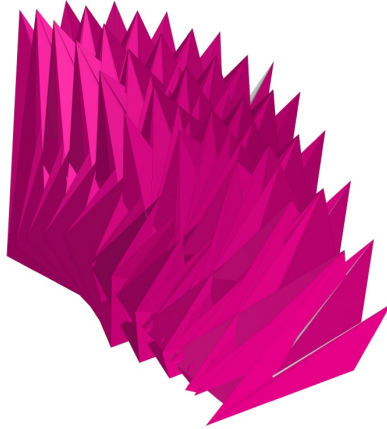


Figure 14

$$1 + 2 + 3 + 4 + 5 + 6 + 5 + 4 + 3 + 2 + 1 = 36 \text{ flaps,}$$

which validates our standard formula for perfectly squared bird bases. Hence, a significant strength of this framework is its accuracy when applied to n^2 bird base compounds within a square. Distinct from box-pleated squares which create exactly ∂^2 for each ∂ on the square, tiled bird bases go further by creating $2\partial+1$ more flaps than the number of squares from the grid.

5 References

- Ghassaei, A. (2017). Origami Simulator. [online] origamisimulator.org. Available at: <https://origamisimulator.org> [Accessed 24 Jun. 2026].
- Lang, R.J., 1996, May. A computational algorithm for origami design. In Proceedings of the twelfth annual symposium on Computational geometry (pp. 98-105).
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